

AI

EK SATHI

YOUR AI COMPANION FOR LEARNING & GROWTH

AI-POWERED STUDENT COMPANION

LEARN • BUILD • GROW

2026–27

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Problem Statement



Students today rely on multiple platforms for learning, project development, career planning, opportunities and technical updates. These activities are scattered across different tools and websites, making the overall student journey fragmented and difficult to manage.

KEY PROBLEMS IN SUMMARY:

- Learning, projects and career activities are handled across multiple platforms
- General AI assistants lack continuous student-specific context and personalization
- GitHub repositories, websites, internships and hackathons require separate searching and understanding
- Useful technology updates are scattered across different sources
- Students need a single platform that connects learning, building, career and opportunities

Motivation & Objectives

ONE PLATFORM. ONE CONTEXT. ONE AI COMPANION FOR THE STUDENT JOURNEY.

MOTIVATION

- Students use multiple disconnected platforms for learning, projects and career growth.
- Useful resources are available, but are not connected to the student's personal goals.
- Existing AI assistants provide general help but limited student-specific continuity.
- We want to build one AI companion that stays relevant to the student's journey.
- The project also gives us an opportunity to combine AI + Cloud + APIs + Data + Personalization in one system.

OBJECTIVES

- Develop a cloud-based AI student companion.
- Provide personalized assistance through a central AI chat.
- Support projects and hackathon preparation.
- Enable GitHub & Website Study through AI-based analysis.
- Provide resume analysis and personalized career roadmaps.
- Discover relevant internships and hackathons.
- Deliver useful technology news and personalized updates.
- Maintain relevant student memory and preferences.

Literature Review, Existing Work & Research Gap

EXISTING WORK / RESEARCH

- **AI in Education**
  Various studies have explored AI for personalized learning, intelligent tutoring and academic assistance.
- **General Purpose AI Assistants**
  Tools like ChatGPT, Gemini etc. provide conversational, coding and general knowledge support.
- **Learning Platforms**
  Platforms like Coursera, Udemy, Khan Academy focus mainly on learning content.
- **Career & Opportunity Platforms**
  Websites like Internshala, LinkedIn, Devpost etc. provide internships, hackathons and career-related opportunities.

OUR PREVIOUS WORK (BOB)

- **BoB – AI Assistant**
  We previously developed BoB, an AI assistant focused on conversational support, basic web analysis and SEO features.
- **Key Learnings**
  We gained practical experience in working with LLMs, building web applications, integrating APIs and deploying a cloud-based system.
- **Limitations**
  BoB was not specifically designed for the complete student journey. It lacked specialized modules for learning, projects, career guidance and opportunity discovery, and did not maintain long-term student context.

RESEARCH GAP & OUR CONTRIBUTION

- **Identified Gap**
  Existing solutions generally address individual aspects of a student's journey. Students still need to use multiple platforms for learning, project development, career planning, opportunities and technical updates. General AI assistants also do not maintain continuous student-specific context.
- **Our Contribution**
  Ek Sathi aims to fill this gap by providing a unified, student-centric AI companion that combines learning, project support, career guidance, opportunity discovery, technical updates and personalized memory in one platform.

01

PROJECT & HACKATHON LAB

Plan projects and prepare smarter for hackathons.

02

GITHUB & WEBSITE STUDY

Decode real projects and websites to learn how they work.

03

CAREER & RESUME

Analyze your resume and build a personalized career roadmap.

04

OPPORTUNITY FINDER

Discover relevant internships, hackathons and opportunities.

Our Solution

EK SATHI

AI-POWERED STUDENT COMPANION

ONE AI COMPANION FOR THE COMPLETE STUDENT JOURNEY

05

TECH NEWS

Stay updated with useful technology news and insights.

06

MY MEMORY

Remember your goals, preferences and important personal context.

USER → CONTEXT & PROCESSING → AI + CLOUD → PERSONALIZED RESPONSE

ONE INTELLIGENT WORKFLOW CONNECTING AI, CLOUD SERVICES AND STUDENT CONTEXT.

How Ek Sathi Works

FROM USER INPUT TO PERSONALIZED AI ASSISTANCE

USER INPUT

ASK • UPLOAD • DISCUSS • SHARE

The student provides a question, document, GitHub repository, website or other resource.

CONTEXT & PROCESSING

NEXT.JS + FASTAPI

The request is processed and relevant user profile, memory and feature-specific context are collected.

AI & CLOUD SERVICES

OPENROUTER + APIS + DATABASE + STORAGE

The LLM processes the request using relevant data from cloud services and external APIs.

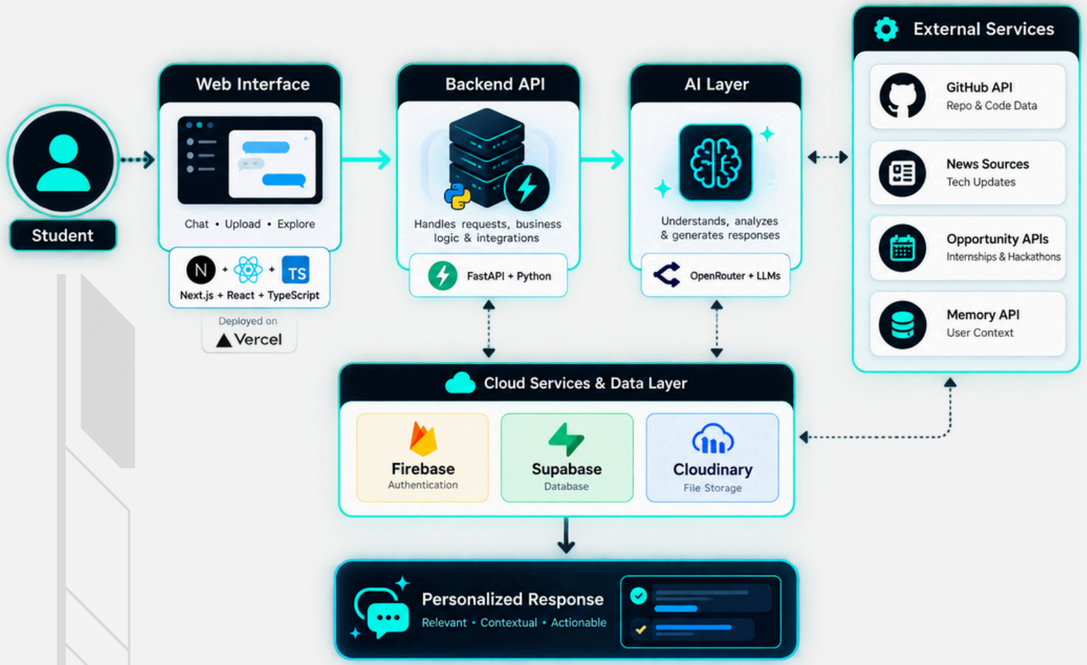
PERSONALIZED RESPONSE

UNDERSTAND • GENERATE • ASSIST

Ek Sathi uses the processed context to provide a relevant and personalized response, analysis or recommendation.

Technology Stack & Cloud Architecture

A CLOUD-BASED ARCHITECTURE CONNECTING AI, DATA & EXTERNAL SERVICES



FRONTEND

Next.js + React + TypeScript

Web interface for chat, dashboard and feature modules.

Deployment: Vercel

BACKEND

FastAPI + Python

Handles APIs, business logic, AI requests and external service **integration.**

AI LAYER

OpenRouter + LLMs

Provides conversational AI, analysis, summarization and personalized assistance.

DATA & AUTHENTICATION

Firebase + Supabase

Firebase manages authentication, while Supabase stores structured application data.

STORAGE

Cloudinary

Manages uploaded files and supported media resources.

EXTERNAL SERVICES

GitHub API + News/Opportunity Sources + Memory API

Provides repository data, technology updates, opportunities and personalized user context.

Societal, Environmental, Sustainability & Ethical Considerations

SOCIETAL IMPACT

- Helps students by bringing learning, projects, career planning, opportunities and technical updates into one platform.
- Reduces the need to search across multiple websites and tools, saving time and effort.
- Supports students from different backgrounds by providing personalized guidance and resources.
- Encourages skill development, project building and participation in internships and hackathons.
- Can help bridge the information gap for students who lack proper guidance or mentorship.

ENVIRONMENTAL & SUSTAINABILITY

- Cloud-based architecture reduces the need for dedicated local hardware or high-end GPUs for individual users.
- Digital access to learning resources reduces dependence on physical materials.
- Modular design allows efficient use of cloud resources and scaling as per demand.
- Promotes long-term reuse and extensibility, reducing redundant development efforts.
- Supports a more accessible and sustainable way of learning and skill development through digital means.

ETHICAL CONSIDERATIONS

- Only publicly accessible and permitted data (e.g., public GitHub repositories and websites) will be processed.
- Private repositories, restricted content, credentials or sensitive information will not be accessed.
- User data, authentication and personal information will be protected and handled securely.
- AI-generated responses and recommendations may not always be perfect; they should be used as assistance, not as the sole basis for important decisions.
- We aim to minimize potential bias by using diverse sources and continuous testing and evaluation.

Work Plan / Development Roadmap

01 | PHASE 1 PLANNING & DESIGN

- Requirement analysis
- Feature finalization
- System architecture design
- Database design
- UI/UX planning

Duration: 2–3 Weeks

KEY DELIVERABLES

- Project requirements
- System design
- UI/UX prototype

02 | PHASE 2 CORE DEVELOPMENT

- Frontend development (Next.js + React)
- Backend development (FastAPI + Python)
- Authentication setup (Firebase)
- Database & storage (Supabase, Cloudinary)
- AI integration (OpenRouter)

Duration: 4–6 Weeks

KEY DELIVERABLES

- Working frontend & backend
- Authentication system
- Database and API integration

03 | PHASE 3 FEATURE MODULES

- Project & Hackathon Lab
- GitHub & Website Study
- Career & Resume Module
- Opportunity Finder
- Tech News Module
- My Memory (user profile & personalization)

Duration: 6–8 Weeks

KEY DELIVERABLES

- All feature modules (MVP)
- Integrated AI assistant
- Personalized user experience

04 | PHASE 4 TESTING & DEPLOYMENT

- Module-wise testing
- AI response evaluation
- Bug fixing & optimization
- Security and data handling checks
- Deployment (Vercel / Cloud)
- User feedback and evaluation

Duration: 3–4 Weeks

KEY DELIVERABLES

- Deployed application
- Testing and documentation
- Initial user feedback

Team Contribution

COLLABORATING TO BUILD EK SATHI

NIKHIL RAWAT
GROUP LEAD

- Overall project planning and coordination
- Requirement analysis and system design
- Backend development (FastAPI + Python)
- AI integration (OpenRouter + LLM APIs)
- Feature development and module integration
- Deployment and cloud configuration (Vercel)
- Documentation and project report

PARTH SINGH
FRONTEND & UI/UX

- Frontend development (Next.js + React + TypeScript)
- UI/UX design and interface development
- Dashboard and feature interfaces
- Responsive design and usability improvements
- Integration with backend APIs
- Testing and bug fixing (UI side)

NISHANT NAGAR
DATABASE & INTEGRATION

- Database design and management (Supabase)
- Authentication setup (Firebase)
- API integration with external services (GitHub, news, opportunity etc.)
- Data processing and storage handling
- Backend support and API testing
- Assistance in deployment and monitoring

NIKHIL DEV SINGH
AI FEATURES & TESTING

- Development of AI feature modules
- Integration with OpenRouter and LLM APIs
- GitHub & website analysis module
- Career, resume and opportunity modules
- Testing, evaluation and performance improvements
- User feedback and content structuring
- Final review and presentation support

Expected Outcomes

FROM AN AI ASSISTANT TO A COMPLETE STUDENT COMPANION

01

PERSONALIZED AI ASSISTANCE

Context-aware support based on the student's goals, preferences and memory.

02

UNIFIED STUDENT PLATFORM

Learning, projects, career, opportunities and technical updates in one place.

03

SMART PROJECT & LEARNING SUPPORT

AI-assisted project development, hackathon preparation and technical study.

04

CAREER & OPPORTUNITY GUIDANCE

Resume analysis, career roadmaps and relevant internship/hackathon discovery.

05

TECHNICAL KNOWLEDGE HUB

AI-processed technology news with the ability to discuss selected topics further.

06

CLOUD-BASED AI ECOSYSTEM

Integrated AI, authentication, database, storage and external services in a scalable web platform.

A SINGLE PLATFORM DESIGNED TO HELP STUDENTS LEARN, BUILD, PREPARE AND GROW.

Future Scope & Conclusion

BUILT FOR TODAY. DESIGNED TO GROW WITH THE STUDENT.

01 **ADVANCED PERSONALIZATION**
Improve AI personalization using richer user context, preferences and long-term memory.

02 **SMART SCHEDULING & PRODUCTIVITY**
Expand chat-based planning with calendar integration, reminders and adaptive study schedules.

03 **INTELLIGENT CAREER SUPPORT**
Add deeper interview preparation, skill-gap analysis and role-specific career guidance.

04 **WIDER OPPORTUNITY INTELLIGENCE**
Expand internship, hackathon and student opportunity sources with smarter recommendations.

05 **ADVANCED LEARNING & CODE ANALYSIS**
Improve repository, website and technical-content analysis for deeper interactive learning.

06 **MULTI-PLATFORM EXPANSION**
Extend Ek Sathi beyond the web to mobile and other student-focused interfaces.

EK SATHI

Your AI Companion for the Student Journey

Learn. Build. Prepare. Grow.

LEARN | BUILD | PREPARE | GROW

A cloud-based ecosystem bringing AI, personalization, learning, career and opportunities together in **one platform**.



Empowering students with AI.
Building a smarter tomorrow, together.

THANK YOU

FOR YOUR TIME & ATTENTION